

Draw the tangent AD , the straight line that just grazes the curve at A , matching its direction there and extending both ways. Between the tangent and the chord opens the angle BAD . As B approaches A , the chord swivels toward the tangent, and this angle is diminished *in infinitum*. For if the angle did *not* vanish, the arc would meet the tangent at a finite rectilinear angle — a sharp elbow at A — and the curvature there would not be continuous, against the supposition. Hence ultimately $\angle BAD \rightarrow 0$. *Q.E.D.*